University of Hertfordshire UH

Logbook 2

Name:	**Jovian Lim Zheng Yuan**
Week Date:	**Part A -Meeting 2 of 8**

(*The above information is for you to keep track of which week this log is applicable for. This information is not required in the online logbook as the system will find your name, your supervisor's name, and take the submission date, automatically. You must submit the online log in the relevant week. Logs* ***cannot*** *be backdated to missed weeks.)*

Meeting date: *(please note if no meeting is planned this week)*	**Date and day** **9 June 2026, Tuesday**
Work Undertaken this week: *Not more than 500 words*	**From the last meeting until now, what have you done based on the task you have listed in the last logbook?** I have updated my hardware block diagram and system diagram. I have added more quality to take note of during my project time. I have finalise on using Arduino uno instead arduino mega since I don't need that much pins. And for my slam programming I will use python instead of C++ as its easier to navigate. For Raspberry pi im using 5 cause it can process Slam Smoothly plus it has it in build in fan to cool the processor down.updated the quality and regulatory.
Problems encountered: *(Not more than 500 words*	**What are the tasks that are stil l outstanding to write your Project Outline** Other then the ones I mention in the upcoming task, I have to check how to train the CNN model for the camera to pick up the different dust, plus I have to research on how the python files will link together. Plus I have to see how are the components are connected to which pin on the board
Tasks agreed for upcoming week before the next meeting	**What are your plan actions/task until the next progress meeting that are agreed with your supervisor** ● **Research on how the dc motor is use as a suction for the vacuum** ● **Research more on Slam integration** ● **Research on how the body of the vacuum should looks like, the weight, dimensions connections if possible** ● **Research on how the python files links together**
Figures: *(Upload image files in the online Log Form, but feel free to keep notes here about which images you intend to upload. Use "Figure Caption"*	**1) Update Hardware Block diagram of your system**

information as the filenames, e.g., "Figure 1 – A screenshot of the initial constrained Sketch in CATIA.jpg")

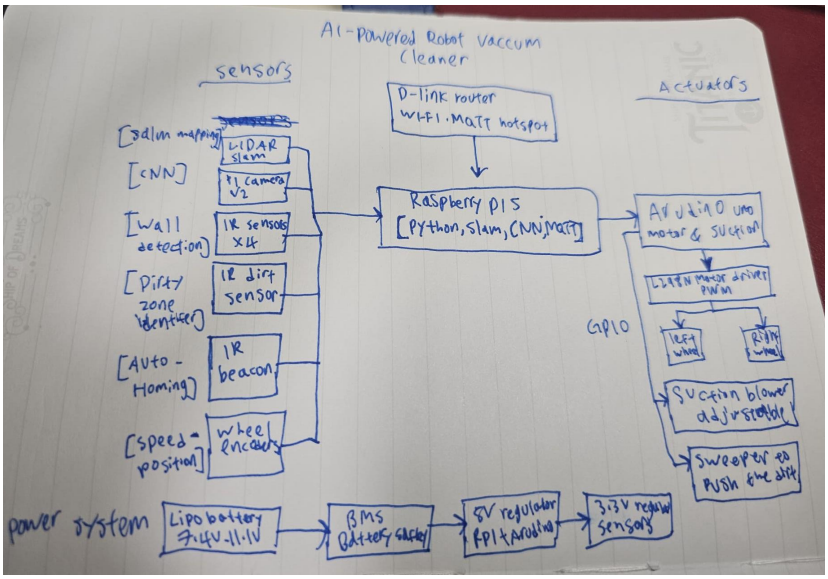


Figure 1: Updated block diagram

2) Key software codes/modules that need to be developed

There will be multiple codes to integrate into this project like such as to call out the hardware, navigation, ai logical thinking and for app communication plus some are utilities code for the backend. Below I

have split them to groups

- Hardware - lidar.py its to call the lidar out to get the distance and views - motor.py control the motor speed and direction -camera.py calls the camerause for cnn -beacon.py for homing when it needs to recharge or go home - Sensors.py reads the sensors, battery voltage, obstacle avoidance -odmetry.py mainly for reading the encoder cycles and show the axis
of the robot
- Navigation -Slam.py updated the map from the lidar scanning plus odmetry -path-planner.py it generates the path to clean from the map -Obstacle Avoidance.py for wall avoidance -controller.py to track the robot between point to point

- AI aspects
  -surface_classifier.py identify the tiles on the floor appearance like
carpet or wood or others
  -dirt_detector.py sees dirt/dust on maps and marks as dirt section
  -suction_control.py controls the power of the motor based on the
floor type

-train_classifier.py trains the model

• App codes
- Server.py for map logging - Mqtt_cilent.py for map updates - Map-encoders to convert numpy to json file for display - Session_logger.py save all logs and history of clean history
• Utilities
- Visualizer.py display the map on the app - Logger.py in the terminal - Simulator.py testing the programme to see hows the lidar work
with - Arduino_flasher.py communicate with raspberry

3) Any Quality and Regulatory issues addressed?

Quality:

1. mechanical safety since theres moving parts like brushes and roller guard, finger or pet wont get caught in between
2. Electrical safety to protect short circuit, over charge and cell balancing protection for BMS, proper wire gauges to prevent wire heatings, Lipo battery casing so it can be kept safety, flame retardant
3.Thermal and overheating management: lipo battery may get overcharge so adding a battery management system will cut the power before it overheats. Include temperature sensors to monitor the motor heat, and shut it down when is too hot.

Regulatory:

1. Product safety, indicating the safety mark for consumer protection , since vacuum uses the main power hence is needed to add
2. Camera/ data privacy : PDPA (personal data protection act 2012) the camera captures the house hence this has the be address since is use for mapping and all, plus people can get captured in the camera
3. Battery fire safety act SS 638 , I will be using a lipo battery and need to be properly labeled so it wont cause others issue

4) A updated/improved version of your system sketch

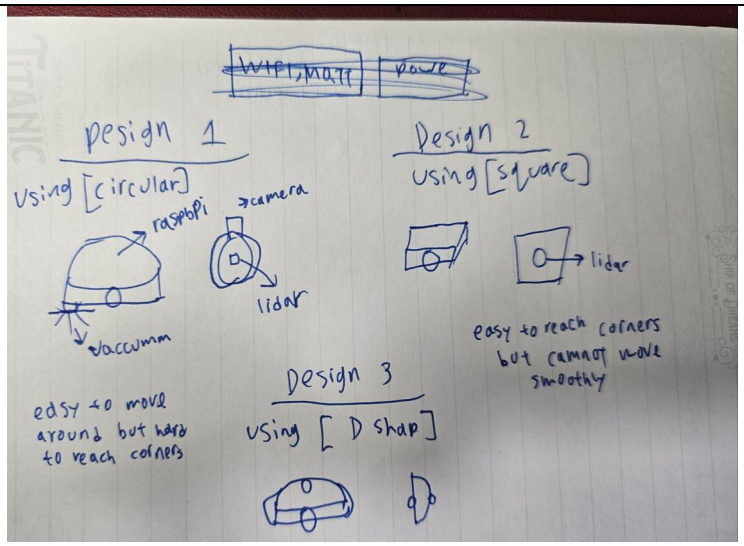


Figure 2: Updated system sketch

References

IEC 60335-2-2 — Household vacuum cleaners (electrical safety)

IEC 62133 — Lithium battery safety

ISO 10218 / ISO 13482 — Robotics safety (ISO 13482 covers personal care robots, very relevant for a home robot)

FCC / CE marking — For Wi-Fi/Bluetooth modules used in the robot